

Resource G Algorithm A (Python).

Import package numpy (for numerical Python) which contains useful mathematical functions.

```
import numpy as np
```

Moisture in alfalfa data is stored in a numerical array. This conversion is useful for applying numpy functions.

```
Z = np.array([7.59, 7.79, 7.81, 7.81, 7.84, 7.86, 7.92,
7.96, 7.99, 8.06, 8.07, 8.07, 8.1, 8.11, 8.15, 8.15,
8.15, 8.16, 8.17, 8.18, 8.19, 8.2, 8.21, 8.22, 8.23,
8.27, 8.27, 8.27, 8.28, 8.29, 8.29, 8.3, 8.31, 8.31,
8.36, 8.57])
```

Compute the classic parameters for comparison with robust.

```
print("Classic")
print(np.mean(Z), np.std(Z))
```

The median of the observed values Z_i is used to obtain the value `max_dist` that is used to detect outliers. It is the median of the absolute deviates to the median multiplied by the correction factor 1.483. This intensive use of the median is frequent for robust estimators. The correction factors were determined by simulation [3, 4].

```
Z_median = np.median(Z)
max_dist = 1.483*np.median(abs(Z - Z_median))
```

Select the convergence criterion. This choice is important whereas, another value may modify the convergence speed.

```
tol = 1e-5
```

The initial value of X , s_A^* and the number of iterations is setup to 0

```
X_robust = 0
sA_rob = 0
temp_val = 0
nb_iterations = 0
```

Start iteration loop. If the difference between the old median of the Z_i and the new one is above the convergence criterion the loop is pursued.

```
while (abs(Z_median - temp_val) > tol):
```

Update correction criterion. It is used to compute the lower or upper distance from the median that is acceptable. Once more, by simulation it was demonstrated that it must be multiplied by 1.5.

```
phi = 1.5*max_dist
```

(Continued)

At each step, each Z_i value is checked. If it too far above or below the median it is replaced by the acceptable extremum. In that way a new set of Z_i is generated.

```
For I in range(len(Z)):
```

The values that are too low or too high are not deleted but replaced by new ones.

```
    if Z[i] < Z_median - phi:
        Z[i] = Z_median - phi
    elif Z[i] > Z_median + phi:
        Z[i] = Z_median + phi
```

Update the robust reference value with the mean of modified dataset.

```
X_robust = np.mean(Z)
```

Update the robust standard deviation with the standard deviation of the modified dataset

```
sA_rob = 1.134* np.std(Z)
```

Swap old values with new and increment the number of iterations.

```
Temp_val = Z_median
Z_median = X_robust
max_dist = sA_rob
nb_iterations = nb_iterations + 1
```

Exit of the loop with printing the robust parameters

```
print("Robust")
X_uncertainty = 1.23*sA_star/np.sqrt(len(Z))
print(X_robust, sA_rob, X_uncertainty)
```

Table 4.6 presents the values obtained for the assigned value X , its standard-uncertainty, and the standard-deviation of the test s_A . Given the type of material analyzed, prepared from a natural sample of unknown content, the most suitable mode of expression for the assigned value, its uncertainty and the standard deviation of the test is given by Algorithm A. For comparison, the values obtained by the classic calculation method are also shown. Table 4.6 shows the role of Algorithm A in modifying the reference value and the standard deviation of the PTS. The major consequence is the reduction of the standard deviation. Both these statistics are used to obtain the scores of the participants and to evaluate their competence. Using the data in Table 4.5, two z -scores are computed, one with the classic parameters, the other with the parameters provided by algorithm A.

Figure 4.3 illustrates the results. The legend gives the meaning of each symbol. Individually measured moisture values appear as circles with the axis on the left. Mean values are reported as assigned references. They are slightly different because they are computed by two methods, classic and robust. The score values are read on the right axis. The red diamonds are used for the classic z -scores, and the green squares for the robust ones. For ease of interpretation of the scores, two horizontal acceptance limits are added at ± 3.0 units (right axis). Classic scores are more tolerant

Table 4.6 Assigned value, standard deviation and uncertainty calculated by different methods.

Method	Reference value X	Standard uncertainty $u(X)$	Standard deviation s_A
Classic	8.12528	Not applicable	0.19622
Algorithm A	8.14512	0.03271	0.15956

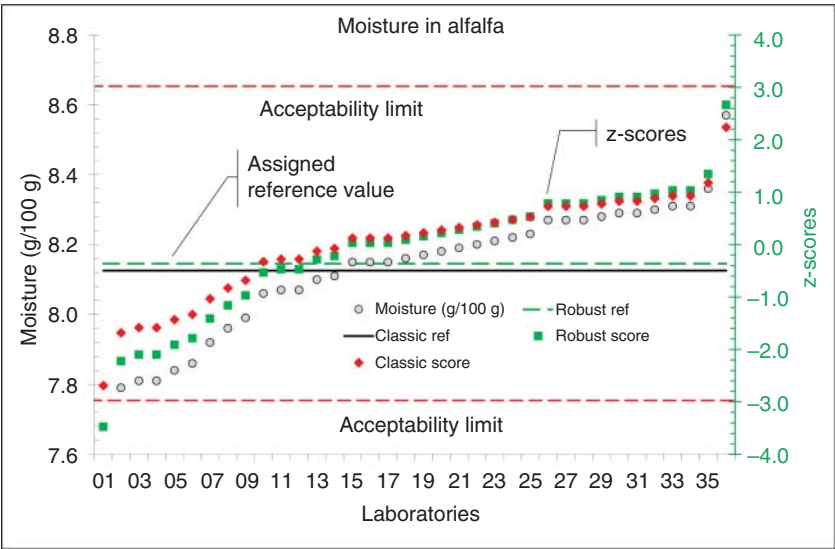


Figure 4.3 ALFALFA – z-scores are reported on the right axis and computed with classic (red diamonds) or robust (green squares) methods. Gray circles are moisture results (left axis).

than robust ones, as no laboratory would be marked as unacceptable, while it is not for one participant with the robust score. The use of PTS results to estimate test sample MU is explained in Section 7.4.1 after the presentation of the general procedure recommended in the GUM.

4.3.6 Check Material Homogeneity or Stability

It is obvious that the material selected for a PTS must be homogenous. Considering the regulatory importance of the latter, a laboratory must not get a bad score because it received an imperfect sample. It may therefore seem paradoxical that ISO 13528 proposes to perform homogeneity checking *after* the completion of the test rather than before. The reason given is that it is first necessary to have the reference value of the material X and the standard deviation of the test s_A to apply the recommended statistical control method for homogeneity checking. Homogeneity can be a difficult goal to achieve. Thus, depending on the type of matrix and the nature of the

analyte, it can be difficult to prepare and maintain an homogeneous test material. Homogeneity checking is often achieved at the end of the manufacturing process, while all test samples are ready to be sent. Considering the overall cost of a PTS, test material manufacturing is the most expensive step. For the organizer, finding out afterwards that the test material is not homogeneous remains the worst situation. On the other hand, for the participants, the lack of homogeneity is a real nuisance because the score they obtained may be unacceptable to an auditor. It is therefore understandable how much care the organizer must take in preparing the material and how crucial it is to check it. To do so, ISO 13528 [5] describes a standardized procedure summarized as follows:

1. Choose a laboratory to conduct the homogeneity check measures and the analytical method to use. If it is acceptable not to perform homogeneity check for each required measurand, select the characteristic or characteristics of the material that are most sensitive to heterogeneity, such as moisture in a food.
2. Prepare and package the proficiency test items for a round of the proficiency testing scheme, ensuring that there are sufficient items for the participants in the scheme and for the homogeneity check.
3. Randomly select I test samples, with $I \geq 10$, in their final packaged form.
4. Prepare J test portions, with $J \geq 2$ from each test item using appropriate techniques to minimize between-test-portion differences.
5. Taking the IJ test portions in a random order, obtain a measurement result Z_{ij} (with $1 \leq i \leq I$ and $1 \leq j \leq J$) on each, completing the whole series of measurements under repeatability conditions.
6. Apply a one-way random analysis of variance (ANOVA) as shown in Section 3.2 and Resource E to calculate the grand mean $\bar{\bar{Z}}$, within-samples standard deviation s_W , and between-samples standard deviation s_B .

The interpretation of the standard deviation depends on the organization of the PTS. But one simple rule is to compare the between-samples standard deviation s_B to the test standard deviation s_A obtained at the end of the PTS. It can be concluded that the test material is homogeneous if:

Acceptance criterion for the test standard deviation

$$s_B \leq 0.3 \times s_A \quad (4.6)$$

Acceptance criterion for the test variance

$$s_B^2 \leq 0.09 \times s_A^2 \quad (4.7)$$

If the variance is considered rather than the standard deviation, this means that the between-samples variance must be less than 9% of the test variance. This is a rule of thumb, and more accurate statistical tests could be applied. In Section 8.3.2, about sampling uncertainty, it is also proposed to use the same experimental design, recommended in an international standard. Computation is simple to obtain for both variances involved in this criterion, as it can easily be achieved with the statistical tools described in Section 3.2.1. Homogeneity checking is a costly requirement since it relies on many measurements. A less expensive method has been proposed

that is based on a non-destructive analytical technique, identified as *rapid* such as near infrared spectrometry (NIRS). NIRS allows a multidimensional approach to homogeneity control, and its ability to highlight small changes in composition that can take place during the manufacturing process is remarkably effective. For this purpose, the entire NIRS spectrum is directly used without having to go through a prediction stage for the constituents [6, 7]. It is also possible to use other statistical techniques based on time series studies, where samples are collected along the manufacturing process. One method of choice is a time series analysis technique called auto-regressive moving average (ARIMA), described by Box and Jenkins [8]. The calculation involved in this method is cumbersome which explains why it was not chosen, although it is extremely sensitive to the smallest variations and allows the detection of slight changes in the spatial distribution of the analyte during the production process.

4.4 Control Charts

Another method used to verify the trueness of measurement results is the implementation of control charts. A very convenient way is to use an RM, certified or not, to develop the control chart. Section 7.3 gives more detail on the distinct types of control charts applicable to controlling a production process. The need for a QC system, such as a control chart, can be simple if the analytical method is accurately developed. As stated in ISO 9000 standards, it is not enough to achieve quality, it is also necessary to have procedures to maintain quality over time. Control charts are adequate tools for the purpose. They are conceptually simple: an RM of known concentration T (for Target) is regularly analyzed. Analyzed test portions are often called quality controls or QCs. The measurement values obtained are plotted, in the chronological order of their collection, on a graph called control chart. If non-random variations around the reference value are visually detected, it is assumed that the measurement system is disturbed, and a corrective action must be taken. To simplify detection, two control limits are drawn on each side of the reference value. It is expected that the responses will remain within these limits. In addition, several rules are defined to decide whether the distribution is no longer random and what corrective action is to be implemented to return to a normal situation. Figure 4.4 provides an example of a control chart for Kjeldahl nitrogen determination in wheat flour. The center line represents the reference target value T of the IRM used for this chart. The outer dashed lines symbolize lower and upper control limits LCL and UCL , and two other dashed lines symbolize lower and upper warning limits LWL and UWL . Warning lines are not always added to the chart.

This example illustrates how an analytical method can diverge from the statistical process control (SPC) required in quality standards. Three typical nonconforming situations are identified. They illustrate the need to take predefined corrective actions, such as recalibration of the measurement system, which is symbolized by the breaks in the curve in the graph: